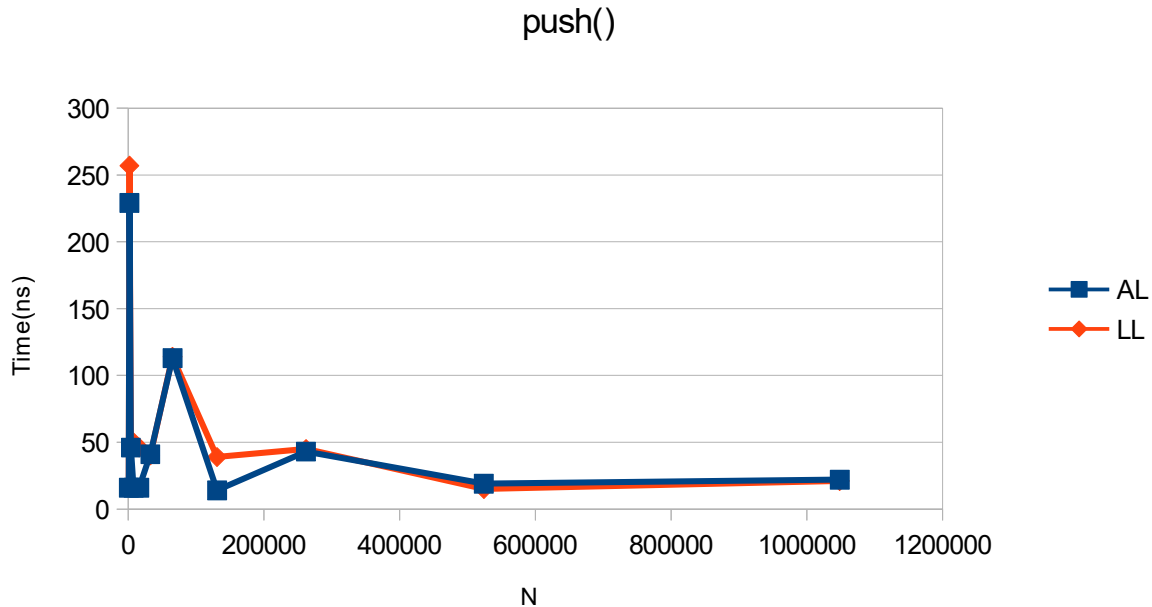


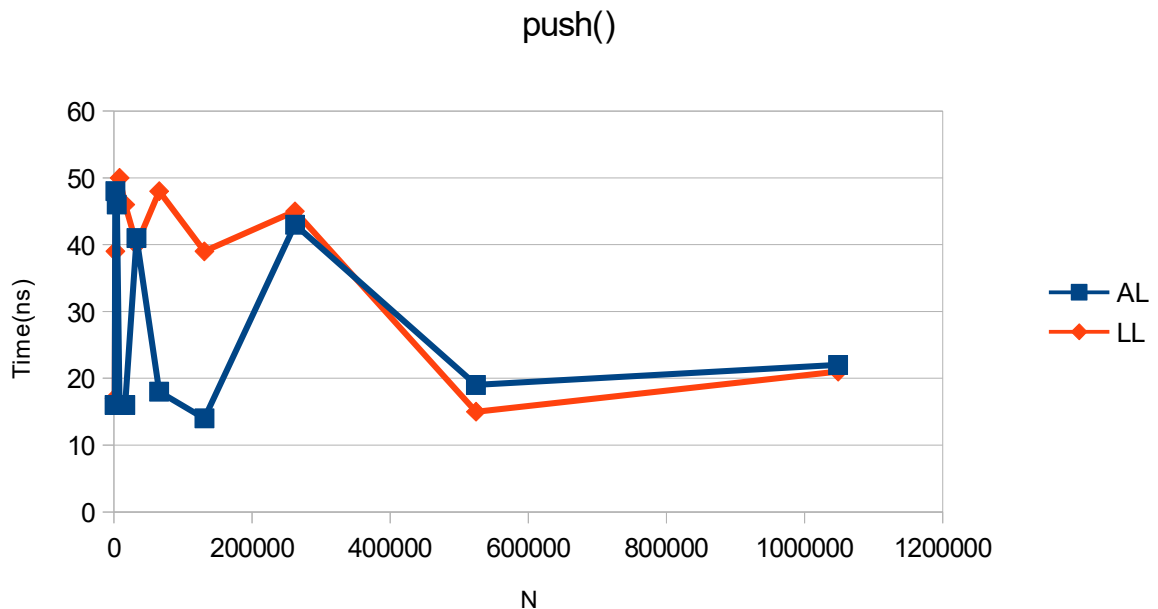
Assignment 5 - Linked List Stack, Analysis - Jon Pulsipher

1) Compare the running time of the push method. What is the growth rate of the method's running time for each stack class, and why?

A: push() is showing $O(1)$ behavior, not growing with N for the ArrayList or Linked List. There are some surprising jumps at $n = 2^{11}$ and $n = 2^{16}$, but they disappear

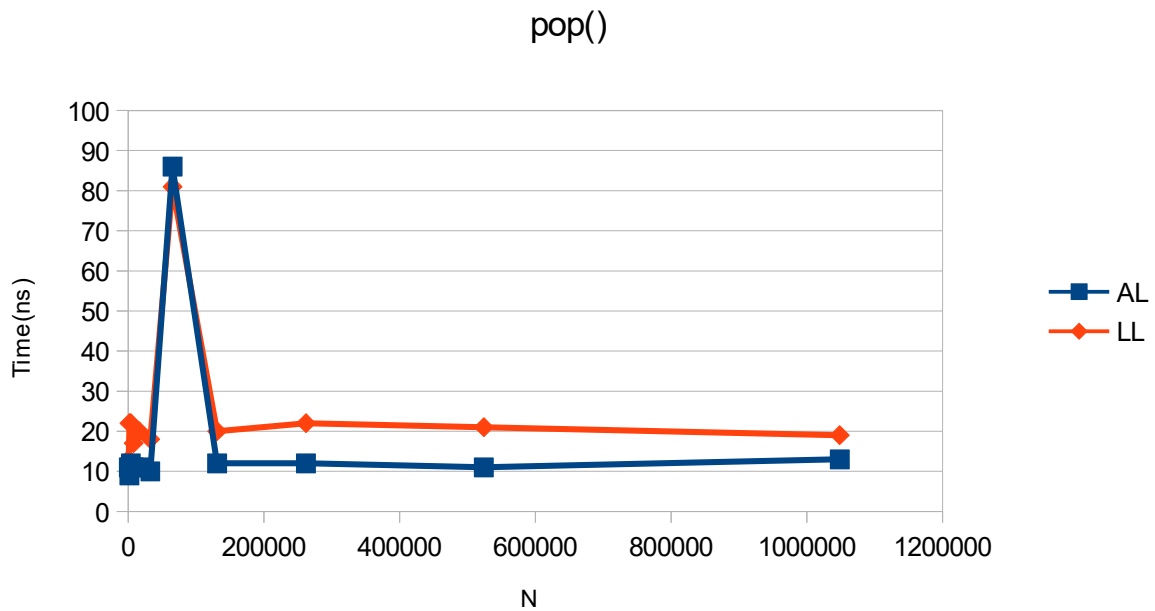


Repeated tests at n^{11} and n^{16} can give a smoother graph (note y-axis range).



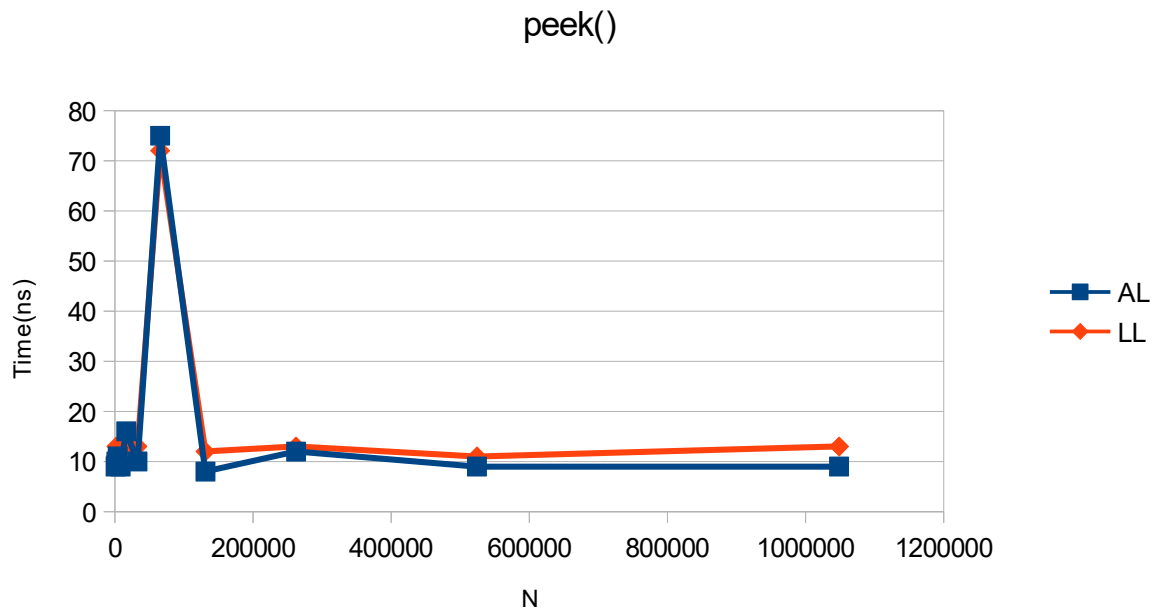
2) Compare the running time of the pop method. What is the growth rate of the method's running time for each stack class, and why?

A: They are very similar, showing $O(1)$ behavior for both, but LL has a higher constant.



3) Compare the running time of the peek method. What is the growth rate of the method's running time for each stack class, and why?

A: $O(1)$ behavior for both again, and LL has a higher constant. again.



4) Based on your timing experiments, which stack class do you think is more efficient for using in your WebBrowser application? Why?

A: I would use the ArrayList stack because of its slight edge on LinkedList shown in pop() and peek().